

**Research and Development for
Space Data System Standards**

**CUSTODY TRANSFER AND
COMPRESSED BUNDLE
STATUS REPORTING**

DRAFT EXPERIMENTAL SPECIFICATION

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DRAFT EXPERIMENTAL SPECIFICATION FOR
CUSTODY TRANSFER AND COMPRESSED BUNDLE STATUS REPORTING

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PREFACE

This document is a CCSDS Experimental Specification. Its Experimental status indicates that it is part of a research or development effort based on prospective requirements, and as such it is not considered a Standards Track document. Experimental Specifications are intended to demonstrate technical feasibility in anticipation of a ‘hard’ requirement that has not yet emerged. Experimental work may be rapidly transferred onto the Standards Track should a hard requirement emerge in the future.

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1 INTRODUCTION

1.1 PURPOSE

The purpose of this document is to define the Compressed Custody Signaling and Compressed Status Reporting extensions for Bundle Protocol version 7.

1.2 SCOPE

This Experimental Specification defines necessary structures and procedures to implement custody transfer and compressed status reporting for Bundle Protocol version 7. It further gives recommendations to implementers on how certain mechanics could be realized and which use cases are enabled by these extensions.

1.3 APPLICABILITY

The specification is applicable to DTN networks based on Bundle Protocol version 7 where custody transfer or compressed status reporting should be available. It is noted that these mechanisms do not need to be implemented on all DTN nodes as these blocks can be forwarded transparently. However, in such scenarios, these mechanisms become less efficient, and users should consider the effects of non-supporting nodes carefully.

1.4 RATIONALE

Regular Bundle Status Reporting is generally discouraged, due to its impact on bandwidth, as each requested report results in a new bundle. Compressed Bundle Status Reporting addresses this issue by reporting on multiple bundles in a single Compressed Bundle Status Report.

Custody Transfer was a feature in Bundle Protocol version 6 [F1] but has been removed for version 7. Still, it is considered a useful feature by many missions and can be used to optimize bundle transfer by moving the responsibility for a certain bundle from one node to the next. There have been some problems associated with custody transfer, such as the estimation of retransmission values or the question of when to take custody. This experimental specification acknowledges these issues but does not aim at providing a complete solution for large-scale, open Disruption Tolerant Networks. It rather seeks to provide sufficient flexibility to address those in smaller, tightly managed networks such as expected to be deployed within the next five to ten years.

Both Custody Transfer and Compressed Status Reporting make use of the same underlying approach for bundle sequence identification. This approach, based on sequence numbers, provides an efficient way to refer to sets of bundles and may have applicability beyond this current specification.

1.5 DOCUMENT STRUCTURE

This Experimental Specification is organized as follows:

- Section 2 provides an overview of the overall approach and the underlying concepts.
- Section 3 specifies the General Concepts for Bundle Sequence Identification.

Commented [SB1]: new extension

Commented [SB2]: as defined in the BPv7 specification

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- Section 4 specifies Custody Transfer in terms of the extension block, the custody signal and the associated procedures.
- Section 5 specifies Compressed Bundle reporting in terms of the extension block and the reporting signal and associated behavior.
- Section 6 discusses how End-To-End services can be realized based on Compressed Bundle Reporting.
- Annex A provides the Protocol Implementation Conformance Statement (PICS).
- Annex B contains Security, SANA/IANA and patent considerations.
- Annex C describes prototyping activities supporting this experimental specification.
- Annex D defines BP Element Nomenclature for Custody Transfer and Compressed Bundle Status Reporting.
- Annex E contains CDDL expressions for the extension blocks and administrative records defined in this specification.
- Annex F provides informative references.

Commented [FF3]: To be added.

1.6 REFERENCES

The following publications contain provisions which, through reference in this text, constitute provisions of this Experimental Specification. At the time of publication, the editions indicated were valid. All publications are subject to revision, and users of this Experimental Specification are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS publications.

[1] CCSDS 734.20-O-1, "CCSDS Bundle Protocol Specification". CCSDS Orange Book, April 2025.

2 OVERVIEW

2.1 SEQUENCE NUMBERS

Sequence numbers are included in `extension` blocks and get generated by Sequence Counters, up to a maximum value. A node can create multiple sequences, e.g. one sequence per destination, through multiple Bundle Sequence Counters. These counters are qualified either by a positive integer called Bundle Sequence ID or a destination endpoint ID.

Commented [SB4]: the proposed

2.2 CUSTODY TRANSFER

Custody Transfer is requested by inserting a Custody Transfer Extension block (CTEB) into a bundle. A node receiving a bundle with a CTEB can decide to become the bundle's custodian by sending an acceptance Compressed Custody Signal (CCS) to the current custodian. A custodian is responsible for storing and retransmitting a bundle, until it receives an acceptance signal (or the lifetime of the bundle expires). The signal uses so called Bundle Sequences to indicate custody acceptance or refusal for multiple bundles. A Bundle Sequence essentially refers to intervals of sequence numbers and their source, extracted from the `CTEB` of received bundles. This allows a custodian that receives a CCS to identify bundles it no longer needs to store and retransmit.

Commented [SB5]: as identified in the CTEB

Commented [SB6]: s

2.3 COMPRESSED STATUS REPORTING

Compressed Status Reporting uses a Compressed Reporting Extension block (CREB) for providing sequence numbers and requesting reporting of bundle status such as reception, deletion or delivery. Such reports will be delivered by a Compressed Reporting Signal (CRS). A CREB is inserted into a bundle when a node wants to be informed about the status of that bundle while traversing the network and other nodes. This status information is provided by the receiving node via a CRS that can be sent to the source that inserted the request or an explicitly specified report-to node and can be used to build additional services such as reliable end-to-end delivery or in-sequence bundle delivery based on such reporting.

Commented [SB7]: Does it always go to the bundle's Report-to endpoint unless that endpoint is dtn:none, in which case the default destination is the administrative endpoint of the bundle's source?

2.4 BUNDLE RETRANSMISSION

Unlike Bundle Protocol version 6, the Custody Transfer Mechanism defined in this specification does not prescribe a specific re-transmission mechanism. Depending on the implementation and the specific use cases, re-transmission can for example be based on timers (like Bundle Protocol Version 6 custody), on received custody signals, explicit commands or even on reliable convergence layers. It is noted that the same or similar mechanisms can be used in connection with Compressed Status Reporting, e.g. to enable reliable bundle delivery. Actually, it can make sense to combine custody transfer for efficiency as it enables re-transmission from one custodian to the next with compressed delivery reporting and eventual end-to-end retransmission for full reliability as custody transfer cannot guarantee end-to-end reliability, e.g., if bundles are lost at a custodial node due to storage constraints or other problems.

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2.5 DISCUSSION: FRAGMENTATION

Bundle fragmentation of bundles which request custody or compressed reporting is problematic as the fragments cannot be uniquely identified by the combination of sequence number, sequence ID and scope node ID anymore. Furthermore, currently no mechanism for bundle re-assembly from fragments is defined or even required by RFC 9171 [F2] which only addresses Application Data Unit Reassembly. Therefore, the current version of this specification does not allow fragmentation of bundles which contain any of the specified extension blocks. These rules must be enforced by by setting the 'Bundle must not be fragmented' in the Bundle Processing Control Flags.

Commented [SB8]: Needs more precise specification.

3 GENERAL CONCEPTS FOR BUNDLE SEQUENCE IDENTIFICATION

3.1 INTRODUCTION

This section defines an efficient way to refer to sequences of bundles using sequence numbers provided by sequence counters. These sequence numbers are used in the extension blocks and administrative records defined in this specification. Sequence numbering allows to refer efficiently to multiple bundles in a sequence and can support the detection of gaps in those sequences. However, in a lot of use cases using a single counter per bundle node is not sufficient to address the various situations efficiently. Therefore, multiple sequences can be used per bundle node with each sequence being identified by a sequence ID.

3.2 BUNDLE IDENTIFICATION

NOTE - Mechanisms for bundle identification are used in extension blocks to later refer to a specific bundle, e.g. in administrative records, or to provide services like gap detection or in sequence delivery.

3.2.1 A **Bundle Sequence Counter** shall generate a sequence of positive integer value which starts at zero and is increased by one each time its current value is included as a **Bundle Sequence Number** into an Extension block.

3.2.2 A Bundle Node may maintain multiple Bundle Sequence Counters which shall be identified either by a positive integer **Bundle Sequence ID**, or by a destination endpoint ID.

NOTE - The assignment of Bundle Sequence IDs is left open to mission or network design to allow optimization for different use cases and specific mission scenarios. For example, Bundle Sequence IDs may be set depending on a service number if the IPN naming scheme is used or the node ID of the next hop.

3.2.3 Bundle Sequence Counters must be maintained separately for Custody Signaling and Status Reporting.

3.2.4 The Bundle Sequence ID '0' shall be reserved and indicate that different Bundle Sequence Counters are maintained for each destination endpoint ID.

3.2.5 When a Bundle Sequence Number is included in an extension block, it shall take the current value of a Bundle Sequence Counter which is then increased by one.

3.2.6 If a Bundle Sequence Number is included in an Extension block, the related Bundle Sequence ID, if non-zero, shall be included as well.

3.2.7 The **Block Source Administrative Endpoint ID** shall be the endpoint ID of the administrative element of the Bundle Protocol Agent inserting an extension block including Bundle Sequence Numbers.

3.2.8 If a Bundle Sequence Number is included in an extension block, a **Block Source Administrative Endpoint ID** shall be included as well unless it is the same as the

Commented [SB9]: Where will this indication appear?

Commented [SB10R9]: Ah. When the bundle sequence ID is zero, the applicable bundle sequence counter is the one that has been established for - and identified by - the bundle's destination EID or, maybe better, destination node.

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source node ID in the primary block.

- 3.2.9** If a Bundle Sequence Counter reaches a maximum value of 2^8 , 2^{16} , 2^{32} or 2^{64} it may be reset to zero.

NOTE – This allows processing nodes to detect possible Bundle Sequence Counter wraparounds.

NOTE – The maximum value should be managed in the node's MIB.

- 3.2.10** The maximum value of a Bundle Sequence Counter must be set in such a way that no extension blocks receive the same Bundle Sequence Number during the existence of that extension block and the potential use of that Bundle Sequence Number in Administrative Records.

NOTE – While such maximum values might be straightforward to set in limited mission scenarios, it is expected that maximum values for larger, more heterogenous networks can be based on network policies defining maximum bundle lifetimes and estimated traffic volumes.

NOTE – For Bundle Sequence ID not equal to '0', the combination of Block Source Administrative Endpoint ID, Bundle Sequence ID and Bundle Sequence Number uniquely identifies a bundle during its lifetime in a given context, such as Custody Transfer or Compressed Status Reporting.

For Bundle Sequence ID '0', the combination of Block Source Administrative Endpoint ID, Destination Endpoint ID and Bundle Sequence Number uniquely identifies the bundle during its lifetime in a given context, such as Custody Transfer or Compressed Status Reporting.

3.3 BUNDLE SEQUENCES AND COLLECTIONS

NOTE – Bundles sequences and collections are used to refer to multiple bundles that can be identified by the identification mechanisms above, e.g. in administrative records.

- 3.3.1** A **Bundle Sequence** shall be a CBOR definite-length array of 3 or 4 items:

1. Bundle Sequence ID as defined in 3.2 - CBOR unsigned integer

or

Destination Endpoint ID – CBOR encoded endpoint ID

2. First Sequence Number – CBOR unsigned integer; this indicates the first Bundle Sequence Number of the first sub sequence in the range of bundles the bundle sequence is referring to

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3. Bundle Sequence Range – A Bundle Sequence Range as defined in 3.3.2 below

or

A CBOR unsigned integer if there is only a single sequence to be included, representing the length of that sequence

NOTE – The Bundle Sequence Range specifies sub-sequence of bundles included and excluded from the report such that the First Sequence Number field of the Bundle Sequence serves as the starting point for the overall range. The Bundle Sequence Range field alternately encodes the lengths of both included and excluded sub sequences.

4. If required: Block Source Administrative Endpoint ID - endpoint ID

Commented [SB11]: CBOR-encoded

NOTE – Whether a Block Source Administrative Endpoint ID is required or not is explicitly stated in the definition of the extension blocks or administrative records using bundle sequences. The Block Source Administrative Endpoint ID is omitted in cases where the receiver of a bundle sequence can implicitly assume the Block Source Administrative Endpoint ID because it is the Administrative Endpoint ID of the receiving node. For example, this is the case for custody signals as defined in Section 4 which never carry the Block Source Administrative Endpoint ID.

NOTE – Later versions of this specification may include additional optional elements, such as status report reason codes as registered in the IANA ‘Bundle Status Report Reason Codes’ sub registry in the ‘Bundle Protocol’ registry or time spans to which reports relate to.

3.3.2 A **Bundle Sequence Range** shall be a CBOR definite-length array with uneven length of CBOR unsigned integers used in bundle sequences to specify included and excluded bundles in the range by giving the length of these sub-sequences.

1. Length of the first bundle sub-sequence included in the bundle sequence
2. Length of the first bundle sub-sequence excluded from the bundle sequence
3. Length of the second bundle sub-sequence included in the bundle sequence
4. Length of the second bundle sub-sequence excluded from the bundle sequence
5. Length of the third bundle sub-sequence included in the bundle sequence
6. ...

NOTE – Bundle sequences report on sub sequences of bundles in the overall range of ‘First Sequence Number – Total Sequence Length -1’. The Total Sequence Length is not explicitly part of a bundle sequence, instead it is the sum of the lengths of all included and excluded sub sequences in that range.

3.3.3 A **Bundle Sequence Collection** shall be a CBOR definite-length array of Bundle Sequence(s).

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~~NOTE – Bundle Sequence Collections may be empty.~~

Commented [FF12]: TBD

NOTE – In contrast to Bundle Sequence Ranges, Bundle Sequence Collections do not need to share the same Bundle Sequence ID or Destination Endpoint ID.

3.4 BUNDLE FRAGMENTATION

- 3.4.1** Extension blocks specified in this specification shall only be added to bundles which have the ‘Bundle Must Not be Fragmented’ flag set or in cases where it can be guaranteed that no fragmentation will occur, e.g. if explicitly forbidden by a network policy.

NOTE – As later fragmentation of bundles using the extension blocks specified in this specification may lead to unintended effects such as loss of expected reporting or custody signals as fragmented bundles are never re-assembled (RfC 9171 does only specify Application Data Unit Re-Assembly), it is recommended to set the ‘Bundle Must Not be Fragmented’ flag at the source if the application of custody transfer or compressed status reporting according to this specification is intended.

4 CUSTODY TRANSFER

4.1 CUSTODY TRANSFER EXTENSION BLOCK (CTEB)

- 4.1.1** A Custody Transfer Extension block shall follow the canonical bundle block format according to RFC 9171, Section 4.3.2.
- 4.1.2** The Block type code shall be `<xx>`.
- 4.1.3** A bundle shall never contain more than one Custody Transfer Extension block.
- 4.1.4** The Block-type-specific data shall be a definite-length CBOR byte string containing a definite-length CBOR array of 3 elements. It shall contain:
- Bundle Sequence Number – CBOR unsigned integer according to 3.2.
 - Bundle Sequence ID – CBOR unsigned integer according to 3.2.
 - Block Source Administrative Endpoint ID - endpoint ID according to 3.2.

Commented [FF13]: Can we ask IANA to assign something between 13 and 191 or do we have to use to have the private/experimental use range 192-255 and have to change for a Blue Book or RFC later?
How would values from the experimental range be coordinated?
(We also need block types for QoS; we need admin record types for BIBE as well)

Commented [SB14]: Why? Why not simply the array?

4.2 COMPRESSED CUSTODY SIGNAL (CCS)

- 4.2.1** A Compressed Custody Signal shall be an administrative record according to RFC 9171, Section 6.1.
- 4.2.2** The record type code shall be `<xx>`.
- 4.2.3** A Compressed Custody Signal shall contain a CBOR map composed by key/value pairs as record content:
- A key of the map shall be the Disposition Code of the Custody Signal,
 - Custody Acceptance: an unsigned CBOR integer:
 - 1 – Custody Accepted (no further information)
 - Values above 1 – reserved for future use
 - Custody refusal: a negative CBOR integer
 - -1 – custody refused (no further information)
 - Values below -1 – reserved for future use
 - Bundle Sequence Collection as defined in 3.3 and subject to the constraints define in 4.2.4 below as associated value.

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NOTE – Later version of this specification might define further positive values for custody acceptance (eg, ‘custody acceptance – duplicate reception’) or further negative values for custody rejection (eg, ‘custody rejection – bundle forwarded’). Implementations may find it useful to test positive or negative values in general and not specifically for ‘1’ or ‘-1’ to maintain some level of compatibility with such future extensions.

- 4.2.4** The Bundle Sequences in the Bundle Sequence Collections in 4.2.3 above shall never contain a Block Source Administrative Endpoint ID.

NOTE – As the administrative element of the receiving node is the same node which included the CTEB initially, a Block Source Administrative Element Endpoint ID is not required as it is implicitly the Administrative Element Endpoint ID of the receiving node.

4.3 CUSTODY PROCEDURES

- 4.3.1** Any node can request custody transfer by inserting a CTEB into a bundle.

Note: The custody transfer requesting node does not have to be the source node of the bundle.

- 4.3.2** Any node receiving a CTEB which is not the destination of a received bundle shall decide whether to take or refuse custody of the received bundle.

NOTE – The decision whether to take custody are mission or network-specific:

- A node may take custody of a bundle when there are ‘reasonably good chances’ that the bundle can be forwarded to a next custodian. This can be based e.g. on a combination of bundle time-to-live, knowledge about upcoming contacts, past experiences, etc.
- By local policy (which could be set by network management)
- Other means

In any case, it is recommended that the conditions under which a node takes or refuses custody are clearly documented.

- 4.3.3** A node receiving a CTEB which is delivering the received bundle shall always take custody of the received bundle.

- 4.3.4** In case the node takes custody, it shall:

- a) store all required information for creating a compressed custody acceptance signal
- b) delete the current CTEB
- c) before forwarding the bundle, add a new CTEB requesting custody transfer with its own Administrative Endpoint ID as the Block Source Administrative Endpoint ID
- d) store the bundle until a custody acceptance signal is received or the bundle life-time

Commented [SB15]: What if the receiving/destination/delivering node does not implement custody transfer?

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has expired.

NOTE – To get informed about any bundle deletion that occurred after custody acceptance, compressed bundle deletion reports can be requested.

- 4.3.5** In case the custodian of a bundle receives a duplicate of a bundle it must (re)-send an acceptance signal and discard the duplicate.

Commented [SB16]: Provided the received bundle itself has a CTEB.

NOTE – A node may also maintain a list of bundles it has accepted custody for in the past, but that have since been transferred to another custodian. This enables the node to suppress late arriving retransmissions by resending an acceptance signal and discarding the bundle.

- 4.3.6** In case the node refuses custody it shall store all required information for creating a compressed custody refusal signal.

NOTE – The node refusing custody may just delete the bundle or try to forward it.

- 4.3.7** A Compressed Custody Signal shall be sent based on the stored information according to the MIB configuration or based on external events.

NOTE - The decision when to transmit a Compressed Custody Signal is an implementation matter:

- Implementation may transmit Compressed Custody Signals periodically based on a timer if there are any CTEB requests to respond to.
- Implementations may transmit Compressed Custody Signals depending on size constraints like the overall report bundle size or the number of bundles to report on. In this case, it is recommended that an additional timer is used to transmit a Compressed Custody Signal even if the size limit is not reached but no additional bundles to report on have been received for some time.
- Implementations may transmit Compressed Custody Signals based on external events (e.g., the establishment of a communication link or reception of an explicit command).

- 4.3.7.1** In case no Compressed Custody Signal acceptance signal is received by a custodian for a specific bundle, the bundle should be re-forwarded.

NOTE – The decision when to re-forward a bundle is an implementation decision. It may be based on:

- Timers: If no custody signal for a bundle is received for a certain amount of time, the bundle may be re-forwarded
- Received Custody Signals: if in-sequence reception of bundles and custody signals can be assumed, gaps in the reporting of accepted custody signals can indicate lost bundles and lead to re-forwarding

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- Explicit command: based on an explicit command, all or certain bundles with pending custody acceptance may be re-forwarded
- Other Internal/External events such as the initiation of a contact
- Reported failure to forward the bundle via a Convergence Layer Adapter

5 COMPRESSED BUNDLE REPORTING

5.1 COMPRESSED REPORTING EXTENSION BLOCK (CREB)

5.1.1 GENERAL REQUIREMENTS

5.1.1 A compressed reporting extension block shall follow the canonical bundle block format according to RFC 9171, Section 4.3.2.

5.1.2 The Block type code shall be `<xx>`.

5.1.3 Any bundle may contain one or more Compressed Reporting Extension block inserted by the same or different bundle nodes.

5.1.4 The Block-type-specific data shall be a definite-length CBOR byte string containing a definite-length CBOR array of 1 to 5 elements as defined in the next clauses.

5.1.2 SEQUENCE NUMBER ONLY (ARRAY LENGTH 1)

NOTE – This can be used where a simple sequential numbering of bundles according to the Destination Endpoint ID is required. This can be used for gap detection or in-sequence delivery of bundles at the destination.

5.1.2.1 If the array length is 1, the array element shall contain a Bundle Sequence Number according to 3.2 above.

5.1.2.2 In the case of array length 1, the Block Source Administrative Endpoint ID shall default to the administrative endpoint ID of the source node.

NOTE – This means that the CREB has been added by the bundle source. The administrative endpoint ID of the bundle source node may in this case not be explicitly known by receivers of this extension block. For certain naming schemes, it could be inferred from the Source Node ID contained in the primary block. However, as for array length one no administrative records are to be created and transmitted to the Block Source Administrative Endpoint ID, the endpoint ID of the administrative element does not need to be known for processing the extension block.

5.1.2.3 In the case of array length 1, the Bundle Sequence ID '0' shall be assumed.

NOTE - According to 3.4.1 this means that separate Bundle Sequence Counters per destination endpoint ID are used. This allows a destination node to detect gaps in bundle reception from a specific source node.

Commented [17]: This reads as if the CBOR array should be encoded into a byte string, which is prefixed with another length field... Leading to two length fields.

Commented [18R17]: Block-type-specific data is a definite byte string, say B1, which specifies the number of bytes. This B1 is then a definite-length CBOR array of 1 to 5 elements, as defined by this CREB specification.

Commented [19R17]: Yes, I am afraid we will need the redundant length information to comply with the standard unless we want to use some other format for the array.

Commented [SB20R17]: RFC 9171 section 4.2.8 says "Block-type-specific data in each block (other than the primary block) SHALL be the applicable CBOR encoding of the content of the block." Doesn't have to be a byte string, and in this case I think it shouldn't.

Commented [21]: For arrays of length 1, 2, 3, and 4, what happens with the values not specified? Is their default value defined somewhere in the spec? is it a managed parameter?

I assume you are not conveying them over the link to maximize efficiency.

Commented [22R21]: Yes, the idea is that for shorter length you have some defaults that do not need to be included. I would hope that's clear enough from the text.

5.1.3 SEQUENCE NUMBER WITH SEQUENCE ID (ARRAY LENGTH 2)

NOTE – This can be used where a sequential numbering of bundles is required. Sequence IDs could e.g. be set according to the source endpoint ID, e.g. the service number in case the IPN naming scheme is used, or the next hop ID to allow for gap detection.

5.1.3.1 If the array length is 2, the array elements shall be

- Bundle Sequence Number – CBOR unsigned integer according to 3.2 above
- Bundle Sequence ID – A Bundle Sequence ID as defined in 3.2

5.1.3.2 In the case of array length 2, the Block Source Administrative Endpoint ID shall default to the administrative endpoint ID of the source node.

NOTE – *See note to 5.1.2.2 above.*

5.1.4 REPORTING REQUEST WITH IMPLICIT BLOCK SOURCE ADMINISTRATIVE ENDPOINT ID (ARRAY LENGTH 3)

NOTE – This can be used for requesting compressed status reporting in cases where the bundle's source node is requesting reporting and the source node ID in the primary block is the endpoint ID of the administrative endpoint.

5.1.4.1 If the array length is 3, the array elements shall be

- The items as defined for an array of length 2 in 5.1.3.1
- Status Report Request Flags – CBOR unsigned integer which shall be processed as bit field indicating the following request flag values
 - Bit 0 – reception reporting request
 - Bit 1 – forwarding reporting request
 - Bit 2 – delivery reporting request
 - Bit 3 – deletion reporting request
 - Bit 4- custody acceptance reporting request
 - Bit 5 – custody rejection reporting request
 - Bit 6 to 63 – reserved for future use

NOTE: Custody related reporting is only for informative purposes and shall not be confused with custody signals which trigger the

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behavior defined in Section 4.3. The intention is to provide reporting of custody related behavior to a report-to node different from the current custodian (which is the receiver of all custody signals).

5.1.4.2 In the case of array length 3, the bundle's source node ID in primary block must be an administrative endpoint ID of the source node.

NOTE – This is required to identify the endpoint ID the administrative records containing the reports shall be sent to.

5.1.4.3 In the case of array length 3, the Block Source Administrative Endpoint ID shall be the administrative endpoint ID of the source node as indicated in the primary block.

NOTE – This means that the CREB has been added by the bundle source.

5.1.5 REPORTING REQUEST (ARRAY LENGTH 4)

NOTE – This can be used by any node for requesting compressed status reporting to its own administrative endpoint.

5.1.5.1 If the array length is 4, the array elements shall be:

- The items as defined for an array of length 3 in 5.1.4.1
- Block Source Administrative Endpoint ID - endpoint ID of the administrative endpoint of the node that created the CREB and assigned the Bundle Sequence ID and the Bundle Sequence Number; reports are sent to this node if no report-to EID is specified (see length 5 below).

5.1.6 REPORTING REQUEST WITH REPORT-TO-EID (ARRAY LENGTH 5)

NOTE – This can be used by any node for requesting compressed status reports to a specific report-to endpoint.

5.1.3.5 If the array length is 5, the array elements shall be:

- The items as defined for an array of length 4 in 5.1.5.1
- Report-to Endpoint ID – the endpoint ID to which the reports shall be send to.

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5.2 COMPRESSED REPORTING SIGNAL

5.2.1 A Compressed Reporting Signal shall be an administrative record according to RfC 9171, Section 6.1.

5.2.2 The record type code shall be **<xx>**.

5.2.3 A Compressed Reporting Signal shall contain as record content a CBOR map of:

- Status Report Reason as key - CBOR unsigned integer providing the reporting reason.
 - 0 – reception report
 - 1 – forwarding report
 - 2 – delivery report
 - 3 – deletion report
 - 4 – custody acceptance report
 - 5 – custody rejection report
 - Values above 6 – reserved for future use
- Bundle Sequence Collection as defined in 3.3 and subject to the constraints define in 5.2.4 below as associated value.

5.2.4 The Bundle Sequences in the Bundle Sequence Collections in 5.2.3 above shall only contain a Block Source Administrative Element ID if the report is to be transmitted to a different report-to endpoint than the administrative element of the node which inserted the reporting request (i.e. case 5.1.6 above).

NOTE – If the administrative element of the receiving node is the same node which included the CREB initially, a Block Source Administrative Element Endpoint ID is not required as it is implicitly the Administrative Element Endpoint ID of the receiving node.

5.2.5 The **Compressed Reporting Signal Destination** shall be:

- the Source Node ID in the primary block in case a CREB with a three-element array has been received (5.1.4 above)
- the Block Source Administrative Endpoint ID in case a CREB with a four-element array has been received (5.1.5 above)
- the Report-To-Endpoint in case a CREB with a five-element array has been received (5.1.6 above)

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5.2.6 Reception of a CREB with a three, four or five-element array (5.1.4, 5.1.5 and 5.1.6 above) shall cause the receiving bundle node to store all required information for generating a Compressed Reporting Signal.

NOTE – This information includes the Compressed Reporting Signal Destination, the Sequence Number, the Sequence ID, the Block Source Administrative Element ID and the status report reason.

5.2.7 Compressed Reporting Signals shall be generated and transmitted to a common Compressed Reporting Signal Destination based on the stored information.

NOTE - The decision when to transmit a Compressed Reporting Signal is an implementation matter:

- Implementation may transmit Compressed Reporting Signals periodically based on a timer if there are any CREB requests to respond to.
- Implementations may transmit Compressed Reporting Signals depending on size constraints like the overall report bundle size or the number of bundles to report on. In this case, it is recommended that an additional timer is used to transmit a Compressed Reporting Signal even if the size limit is not reached but no additional bundles to report on have been received for some time.
- Implementations may transmit Compressed Reporting Signals based on external events (e.g., the establishment of a communication link or reception of an explicit command).

6 DISCUSSION: CBR-BASED END-TO-END SERVICES

6.1 END-TO-END SERVICES

This section discusses how the mechanisms introduced in Section 5 can be used to realize end-to-end services between a bundle source and destination. Some of the services described in this section offer similar functionality as provided by the Delay-Tolerant Payload Conditioning (DTPC) Transport Service as defined in CCSDS 734.2-B-1, Annex E, for BPv6 [F1]. In contrast to DTPC which realizes its services as an BP Application Layer Protocol by introducing DTPC PDU to be transmitted in the payload block of a bundle, the services described here rely solely on the use of the extension blocks described in this specification. However, DTPC offers additional features (such as topics) and an aggregation service so that the mechanisms described here do not offer a full replacement for DTPC.

The services described in the following are not mutually exclusive and some or all of them can be combined without the need for additional extension blocks.

The mechanisms described in 6.2 to 6.3 do work with any type of CREB, even with the simple form as defined in 5.1.2, and do not require the generation of any Compressed Reporting Signals.

End-to-end accounting (Section 6.4) and end-to-end reliability (Section 6.5) do require the source node to insert a CREB assigning sequence numbers based on the destination endpoint ID and requesting at least delivery reporting and maybe additionally deletion reporting.

6.2 END-TO-END IN-SEQUENCE DELIVERY

If the sequence numbers contained in the CREB are based on separate Bundle Sequence Counter per destination endpoint ID (ie, Bundle Sequence ID '0'), these sequence numbers can be used to re-order out-of-order bundles at the destination. Some care must be taken to account for any lost bundles or late arrival of bundles. Typically, this could be handled by time outs.

6.3 DUPLICATE SUPPRESSION

Duplicate Suppression works in a similar manner as the In-Sequence Delivery described in Section 6.2: If the sequence numbers contained in the CREB are based on separate Bundle Sequence Counter per destination endpoint ID (ie, Bundle Sequence ID '0'), these sequence numbers can be used to detect duplicated bundles (as they will have received the same sequence number) and suppress them. By relying on sequence numbers, this can be implemented in a much more efficient way than by using bundle time stamps which would require the storage of all timestamps.

6.4 END-TO-END GAP DETECTION

End-to-end Gap Detection can also be based on sequence numbers which are based on separate Bundle Sequence Counter per destination endpoint ID (ie, Bundle Sequence ID '0'). Again, care must be taken for late arrival of bundles which could be based on some sort of time-out.

6.5 END-TO-END ACCOUNTING

The use of compressed status reporting for delivery or deletion of bundles does allow for end-to-end accounting of bundles towards the source node (using the request format in Section 5.1.5) or to a different report-to endpoint ID (using the request format in Section 5.1.6).

While delivery reporting does only require the destination node to support compressed status reporting, deletion accounting does also require the deleting node to support compressed status reporting. If the deleting node does not support compressed status reporting, failed delivery of a bundle can still be assumed by a missing delivery report. However, there may be cases where the reporting signal is lost and thus a delivery failure is wrongly assumed.

6.6 END-TO-END RELIABILITY

Using the end-to-end accounting mechanism described above, end-to-end reliability can be easily provided by re-transmitting a bundle where an expected delivery report has not been received or which has been reported as deleted.

As re-transmitting a bundle again over the full end-to-end path is usually not efficient, this mechanism may be combined with Custody Transfer as specified in Section 4. In this way, re-transmission will typically happen between custodians with end-to-end re-transmission only required in cases where the custody transfer fails.

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ANNEX A
PROTOCOL IMPLEMENTATION CONFORMANCE STATEMENT PROFORMA
(NORMATIVE)

A1 INTRODUCTION

A1.1 GENERAL

This annex provides the Protocol Implementation Conformance Statement (PICS) Requirements List (RL) for implementations of this experimental specification. The PICS for implementation is generated by completing the RL in accordance with the instructions below. An implementation shall satisfy the mandatory conformance requirements of the base standards referenced in the RL.

For this specification four distinct features have been identified:

- Custody Transfer (Annex A4)
- Compressed Reporting – CREB Sequence Numbering (Annex A5)
- Compressed Reporting – CREB Reporting Requests (Annex A6)
- Compressed Reporting – Compressed Reporting Signal (Annex A7)

Implementations may implement any combination of these features but need to clearly state which features are supported. Support of a feature requires implementation of all mandatory requirements for that specific feature.

An implementation's completed RL is called the PICS. The PICS states which capabilities, features and options of the protocol have been implemented. The following can use the PICS:

- the protocol implementer, as a checklist to reduce the risk of failure to conform to the standard through oversight;
- the supplier and acquirer or potential acquirer of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma;
- the user or potential user of the implementation, as a basis for initially checking the possibility of interoperating with another implementation.
- a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A1.2 NOTATION

The following are used in the RL to indicate the status of requirements:

A1.2.1 Status Symbols

- M mandatory.
- O optional.

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Please note that this PICS is organized into features (A4 to A7) and ‘mandatory’ means mandatory if the specific feature is implemented and support is indicated in A3 below.

A1.2.2 Support Column Symbols

The support of every item as claimed by the implementer is stated by entering the appropriate answer (Y, N, or N/A) in the support column:

Y	Yes, supported by the implementation.
N	No, not supported by the implementation.
N/A	Not applicable.

Not supported features shall be indicated as ‘N/A’ in the tables in A4 to A7 below.

A1.3 INSTRUCTIONS FOR COMPLETING THE RL

An implementer shows the extent of compliance to the protocol and its features by completing the RL; that is, the RL shows compliance to all mandatory requirements and lists all options that are not supported. The resulting completed RL is called a PICS. In the Support column, each response shall either be a response code selected from the indicated set of responses or else comprise one or more parameter values, as appropriate. If a conditional requirement is inapplicable, N/A should be used. If a mandatory requirement is not satisfied, exception information must be supplied by entering a reference Xi, where i is a unique identifier to an accompanying rationale for the noncompliance.

A1.4 REFERENCED BASE STANDARDS

The base standards referenced in the RL are:

- Custody Transfer and Compressed Bundle Status Reporting (this document).

A1.5 IDENTIFICATION OF PROTOCOL

Ref	Question	
1	Date of Statement (DD/MM/YYYY)	
2	CCSDS Reference	CCSDS-
3	Protocol Name	Custody Transfer and Compressed Bundle Status Reporting

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A2 IDENTIFICATION OF THE IMPLEMENTATION

Implementation Name	
Implementation Version	
Special Configuration	
Supplier	
Contact Point for Queries	
Other Information Necessary for full identification, for example, Project developed for, name(s) and version(s) for machines and/or operating systems, system name(s).	

A3 IMPLEMENTATION SCOPE & FEATURES

The following features are identified and may be implemented separately. A ‘YES’ implies the applicability of all requirements contained in the referenced section.

Custody Transfer (A4)	YES / NO
Compressed Reporting – CREB Sequence Numbering (A5)	YES / NO
Compressed Reporting – CREB Reporting Requests (A6)	YES / NO
Compressed Reporting – Compressed Reporting Signal (A7)	YES / NO
Have any exceptions within the implementation scope been required? If so, identify the affected feature and the exceptions. (Note: A YES answer means that the implementation of a protocol feature does not conform to the specification. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.)	YES / NO

A4 CUSTODY TRANSFER

Type ID	Requirements	Reference	Status	Support
CT-01	Custody Transfer Extension Block	4.1	M	
CT-02	Compressed Reporting Signal	4.2	M	
CT-03	Custody Procedures	4.3	M	

A5 COMPRESSED REPORTING – CREB SEQUENCE NUMBERING

Type ID	Requirements	Reference	Status	Support
SN-01	Compressed Reporting Extension Block – General Requirements	5.1.1	M	
SN-02	Compressed Reporting Extension Block – Array Length 1	5.1.2	M	

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Type ID	Requirements	Reference	Status	Support
SN-03	Compressed Reporting Extension Block – Array Length 2	5.1.3	M	
SN-04	Compressed Reporting Extension Block – Array Length 4	5.1.4	M	

A6 COMPRESSED REPORTING – CREB REPORTING REQUESTS

Type ID	Requirements	Reference	Status	Support
RR-01	Compressed Reporting Extension Block – General Requirements	5.1.1	M	
RR-02	Compressed Reporting Extension Block – Array Length 3	5.1.5	M	
RR-03	Compressed Reporting Extension Block – Array Length 4	5.1.6	M	

A7 COMPRESSED REPORTING – COMPRESSED REPORTING SIGNAL

Type ID	Requirements	Reference	Status	Support
RS-01	Compressed Reporting Signal	5.2	M	

ANNEX B

SECURITY, SANA/IANA, AND PATENT CONSIDERATION (INFORMATIVE)

B1 SECURITY CONSIDERATIONS

B1.1 INTRODUCTION

In general, any deployment of DTN nodes and use of DTN networks must consider security aspects based on a detailed security risk assessment considering among other aspects the specifics of the network, the involved parties and the criticality of the data being transmitted. It is expected that corresponding security requirements and network security policies will be defined to which participating nodes and their operations must conform to. In the following, some specific aspects related to Custody Transfer and Compressed Bundle Status Reporting are discussed but it is important to note that these should not be considered in isolation but together with the overall network security aspects.

B1.2 SECURITY CONCERNS WITH RESPECT TO THE CCSDS DOCUMENT

The mechanisms specified in this document have been designed with relatively small, well-managed DTN networks in mind where all participating parties are known and agreements on the use of the network and its resources have been established. For security purposes, application of BPSEC [F3] is recommended.

B1.2.1 DATA PRIVACY

Some or all the data which can be reported via compressed bundle status reports may be considered sensitive. To avoid that unauthorized or malicious nodes request reports, the CREB can be authenticated by a BPSEC Bundle Integrity Block by the inserting node so that reporting nodes can check whether the CREB is authentic for the contained Block Source Administrative Endpoint ID and use policies to determine whether reporting to that node or an eventual report-to endpoint id is allowed. The Compressed Reporting Signals can be protected by a BPSEC bundle confidentiality block.

B1.2.2 DATA INTEGRITY

Data Integrity of CTEB and CREB extension blocks can be provided by BPSEC Bundle Integrity Blocks targeting these extension blocks. In a similar way, data integrity of Compressed Custody Signals and Compressed Reporting Signals can be provided by a BPSEC Bundle Integrity Block targeting the payload block of the signaling bundle as these signals are provided as administrative records within the block.

B1.2.3 AUTHENTICATION OF COMMUNICATING ENTITIES

The block sources for CTEB and CREB extension blocks can be authenticated by a BPSEC Bundle Integrity Block. The source of Compressed Custody Signals and Compressed Reporting

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Signals can be authenticated by a BPSEC Bundle Integrity Block targeting the bundle's primary block.

B1.2.4 CONTROL OF ACCESS TO RESOURCES

For Custody Transfer, custody acceptance can be based on the successful authentication of the previous custodian which inserted the CTEB or the bundle's source node. Custody acceptance can be limited to specific previous custodians or bundle sources to protect storage and communication resources.

For Compressed Status Reporting, reporting can be based on the successful authentication of the CREB block source and can be limited to specific block sources or report-to endpoint ID to protect storage and communication resources.

It is noted, however, that the use of BPSEC Integrity Blocks may have an impact on processing resources.

B1.2.5 AVAILABILITY OF RESOURCES

Availability of resources can be protected by the same means as described in B1.2.4.

B1.2.6 AUDITING OF RESOURCE USAGE

It is recommended that the usage of processing, storage and communication resources of bundles nodes is monitored anyway. With respect to Custody Transfer, specific monitoring parameters may include the amount of storage currently being used for bundles under custody, eventually being distinguished by bundle source or destination.

It is also recommended to monitor the total traffic generated due to custody reporting and compressed bundle status reporting.

B1.3 POTENTIAL THREATS AND ATTACK SCENARIOS

Malicious nodes may insert CTEB to require nodes to store bundles for some time. Nodes may protect against this by the authenticating and authorization as described in B1.2.4 above.

Malicious nodes may take custody of a bundle and discard it silently causing bundle loss. For mitigation, compressed delivery reporting can be applied to bundles to detect such losses and take action to prevent further forwarding of bundles to those malicious nodes.

Malicious nodes may also try to flood nodes with compressed reports by inserting those as report-to endpoint IDs. These can be mitigated by the mechanisms described in B1.2.4 above.

B1.4 CONSEQUENCES OF NOT APPLYING SECURITY TO THE TECHNOLOGY

If no security mechanisms are applied, communication, storage and processing resources may become unavailable or constraint due to malicious nodes requesting custody transfer, compressed bundle status reporting or sending false custody refusal signals causing unnecessary retransmissions. Bundles may be lost due to malicious nodes sending false custody acceptance signals and discarding the bundle.

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B2 SANA/IANA CONSIDERATIONS

As no specific resources for block type or administrative record types have been delegated by IANA to SANA those numbers will be requested directly from IANA.

The following registrations will be requested:

B2.1 BUNDLE BLOCK TYPES

<https://www.iana.org/assignments/bundle/bundle.xhtml#block-types>

Bundle Protocol Version	Value	Description	Reference
7	13	Custody Transfer Extension Block	CCSDS-... (this book)
7	14	Compressed Status Reporting Extension Block	CCSDS-... (this book)

B2.2 BUNDLE ADMINISTRATIVE RECORD TYPES

<https://www.iana.org/assignments/bundle/bundle.xhtml#admin-record-types>

Bundle Protocol Version	Value	Description	Reference
7	13	Compressed Custody Signal	CCSDS-... (this book)
7	14	Compressed Reporting Signal	CCSDS-... (this book)

B3 PATENT CONSIDERATIONS

There are no patents that are known to apply to this specification.

ANNEX C PROTOTYPING (INFORMATIVE)

Commented [FF23]: Will be updated a bit.

C1 ESA BUNDLE PROTOCOL IMPLEMENTATION

All features described in this specification have been implemented in ESA's Bundle Protocol Implementation and validated.

To validate the lower-level concepts like sequence number generation and conformance to the structure definitions, unit tests have been employed that assert proper sequence number management as well as correct creation and proper encoding / decoding of structures (Extension Blocks, Administrative Records and Bundle Sequence structures).

The higher-level concepts have been validated through simulations and focus on testing the aspects outlined in C2-C4. The tests focus on a source node adding the extension blocks, with other nodes in the network being configured to test a certain scenario, e.g. by having an artificial low storage capacity to test custody refusal.

In the following a brief description of some implementation aspects is provided.

C1.1 SEQUENCE NUMBER GENERATION

ESA BP implements the Bundle Sequence Counters as two maps (one for Custody Transfer and one for Compressed Bundle Status Reporting) of Sequence Identifiers – a wrapper for either a Destination Endpoint ID or a Bundle Sequence ID - mapped to Sequence Numbers. Each map is accessible through a function that accepts a Sequence Identifier and either inserts a new mapping, increments the current count, or resets the count based on the configured maximum. The updated value is then used for the creation of Extension Blocks.

C1.2 CUSTODY ACCEPTANCE/REFUSAL

ESA BP evaluates whether to take custody of a bundle during reception. The destination node will always accept custody, while intermediate nodes may refuse custody if either the bundle cannot be forwarded (e.g., when no next hop is known) or if a configurable storage threshold is reached.

When custody is accepted, a dedicated retention constraint is added to the bundle that is removed upon reception of an accepting CRS. When custody is refused due to not being able to forward, the bundle will be deleted. If custody is refused due to reaching the storage threshold but the bundle can be forwarded immediately, it will be forwarded and deleted afterwards.

C1.3 RETRANSMISSION OF BUNDLES

Only bundles for which custody has been accepted will be retransmitted. This will either happen after a configurable timeout or upon reception of Compressed Custody Signals. The timeout is realized by storing a dedicated retransmission time for the bundle that is periodically checked against the current time. When a refusing CRS is received, the retransmission time will be reset for all refused bundles to try sending the bundle later, eventually via another node. When an

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accepting CRS is received, all bundles in the corresponding excluded sub-sequences of the received Bundle Sequence Ranges will immediately be retransmitted.

C1.4 GENERATING AND SENDING COMPRESSED SIGNALS

When a bundle reaches certain stages in its lifecycle inside ESA BP, (e.g. after reception or after custody acceptance/refusal), all necessary information to generate a Compressed Signal is stored in what we term a Compressed Item. These items are then stored in a temporary item buffer that is identified by the corresponding signal destination. Whenever a new destination is added, a timer is started that, once reaching a configurable timeout, triggers processing of the items. Additionally, once a buffer contains a certain number of items, processing will be triggered prematurely. During processing, the buffer is removed, items are aggregated into a Compressed Signal, and the signal is sent to its destination.

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ANNEX D
BP ELEMENT NOMENCLATURE FOR
CUSTODY TRANSFER AND COMPRESSED STATUS REPORTING
(INFORMATIVE)

Commented [FF24]: @Simon: Are you going to provide something for this?

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ANNEX E

CDDL EXPRESSIONS
(INFORMATIVE)

For informational purposes, we provide an expression of this specification in the Concise Data Definition Language (CDDL). The CDDL expression is presented below. Note that wherever the CDDL expression disagrees with the textual representation of the specification presented in the earlier sections of this document, the textual representation rules. The definitions below utilize CDDL expressions from RFC 9171.

```
; Compressed Custody Signal
$admin-record /= [xx, custody-signal-content]
custody-signal-content = {
  -1 => seq-collection; Custody rejected
  1 => seq-collection; Custody accepted
  * int => seq-collection; Reserved
}

; Bundle Sequence Collection
seq-collection = [+ seq]
; Bundle Sequence
seq = [
  seq-id-ref: (unit .gt 0) / eid,
  seq-start: seq-num,
  range: seq-range / seq-incl-len,
  ? blk-source
]

; Bundle Sequence Range
seq-range = [seq-incl-len, +(seq-exl-len, seq-incl-len)]
seq-incl-len = uint .gt 0; inclusive length
seq-exl-len = uint .gt 0; exclusive length

; Bundle Sequence Number
seq-num = uint

; Bundle Sequence ID
seq-id = uint

; Block Source Administrative Endpoint ID
blk-source = eid
```

Commented [YJ25]: We don't explicitly disallow empty collections, though it would not make much sense to have them (currently, we are able to parse them and do not throw an exception) empty would use * instead of +

Commented [FF26R25]: Maybe we should allow them. Nodes may periodically send reports and they could be empty if there is nothing to report. I have added a note.

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```
; Compressed Reporting Signal
$admin-record /= [xx, reporting-signal-content]
reporting-signal-content = {
    6 => seq-collection; Custody rejection report
    5 => seq-collection; Custody acceptance report
    4 => seq-collection; Reassembly report
    3 => seq-collection; Deletion report
    2 => seq-collection; Delivery report
    1 => seq-collection; Forwarding report
    0 => seq-collection; Reception report
    * int => seq-collection; reserved
}

; CTEB
$extension-block /= extension-block-use<xx, embedded-cbor<ext-
data-custody-transfer>>
ext-data-custody-transfer = [
    seq-num
    seq-id,
    blk-source
]

; CREB
$extension-block /= extension-block-use<xx, embedded-cbor<ext-
bundle-reporting>>
ext-bundle-reporting /= [
    seq-num,
    ?(
        seq-id,
        ?(
            status-report-request-flags,
            ?(
                blk-source,
                ? report-to-eid: eid
            )
        )
    )
]
]
```


DRAFT EXPERIMENTAL SPECIFICATION FOR
CUSTODY TRANSFER AND COMPRESSED BUNDLE STATUS REPORTING

```
status-report-request-flags = uint .bits statusrequestbits
statusrequestbits = &(
    reserved: 7,
    custody-rejected: 6,
    custody-accepted: 5,
    reassembly: 4,
    deletion: 3,
    delivery: 2,
    forwarding: 1,
    reception: 0
)
```

DRAFT EXPERIMENTAL SPECIFICATION FOR
CUSTODY TRANSFER AND COMPRESSED BUNDLE STATUS REPORTING

ANNEX F
INFORMATIVE REFERENCES
(INFORMATIVE)

[F1] CCSDS 734.2-B-1 , “CCSDS Bundle Protocol Specification”, September 2015.

[F2] Burleigh, S., Fall, K., and Birrane, E. III. “RFC 9171: Bundle Protocol Version 7”. IETF, 2022.

[F3] CCSDS 734.5-R-2 “CCSDS Bundle Protocol Security Specification”. CCSDS Red Book, September 2023.

Commented [27]: Is there something newer which can be referenced?